Transmission-Line Signal Integrity: Recovering Hidden $R(x)$ and $C(x)$ Profiles via TDR Inversion and ℓ_1 Linear Programming

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***Abstract*—Time-domain reflectometry (TDR) serves as an essential non-destructive diagnostic methodology for localizing hidden faults along high-speed transmission lines and printed circuit interconnects. In this work, we reconstruct the non-uniform distributed resistance $R(x)$ and capacitance $C(x)$ profiles of a 1.0 m transmission line from terminal voltage recordings at $x = 0$ following a fast voltage step. Beginning with Maxwell's curl equations, Stokes' theorem is applied over an elemental transmission slice to yield the coupled first-order Telegrapher's system, which is decoupled into a second-order hyperbolic damped wave partial differential equation. Transform-domain Laplace analysis establishes the complex propagation constant and characteristic impedance, explaining the fundamental physical reason why small series resistance faults are unobservable relative to shunt capacitance defects under finite measurement noise. Forward transient wave propagation is simulated with a staggered-grid leapfrog finite-difference time-domain (FDTD) scheme operating under Courant-Friedrichs-Lewy stability. Parameter deviations are subsequently inverted by mapping the TDR boundary misfit through a sensitivity Jacobian into an ℓ_1-norm linear program solved with the HiGHS dual-simplex solver. Four successive linear programming (SLP) passes eliminate first-order linearization error, locating a 40% capacitance defect at $x \approx 0.5$ m to within 0.001 grid cells under 0.1 mV noise and dropping the residual root-mean-square to the 0.09 mV physical noise floor.**

***Index Terms*—Time-Domain Reflectometry, Telegrapher's Equations, Damped Wave Equation, Finite-Difference Time-Domain, ℓ_1 Linear Programming, Successive Linear Programming.**

I. Field Model

To model electromagnetic pulse propagation along a two-conductor transmission line, we consider an infinitesimal line segment extending from spatial coordinate x to $x + \Delta x$. The electromagnetic field distributions in and around the conductors are governed by Maxwell's curl equations, which express Faraday's law of induction and Ampère's circuital law with Maxwell's displacement current correction:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \tag{1}$$

where $\mathbf{E}$ is the electric field intensity, $\mathbf{B}$ is the magnetic flux density, $\mathbf{H}$ is the magnetic field intensity, $\mathbf{J}$ represents the conduction current density, and $\mathbf{D}$ denotes the electric displacement field.

We apply Stokes' theorem to Faraday's law over an oriented planar loop bounded between the two parallel conductors over the longitudinal segment Δx. The circulation of $\mathbf{E}$ along this closed boundary balances the potential difference across the conductors against the ohmic drop and the time rate of change of magnetic flux linking the loop:

$$\oint_{\partial S} \mathbf{E} \cdot d\boldsymbol{\ell} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \tag{2}$$

By defining $V(x,t)$ as the transverse line voltage, $I(x,t)$ as the longitudinal current, $R(x)$ as the per-unit-length distributed series resistance in Ω/m, and $L(x)$ as the per-unit-length distributed series inductance in H/m, taking the limit as $\Delta x \to 0$ yields the first Telegrapher equation:

$$\frac{\partial V(x,t)}{\partial x} = -R(x)I(x,t) - L(x)\frac{\partial I(x,t)}{\partial t} \tag{3}$$

In an analogous manner, we apply Gauss's divergence theorem and charge continuity to an elemental conductor volume of length Δx. The longitudinal divergence of current balances transverse leakage across the dielectric substrate through per-unit-length conductance $G(x)$ in S/m and capacitive displacement current charging the conductors through per-unit-length capacitance $C(x)$ in F/m:

$$\frac{\partial I(x,t)}{\partial x} = -G(x)V(x,t) - C(x)\frac{\partial V(x,t)}{\partial t} \tag{4}$$

Equations (3) and (4) form the coupled first-order system of Telegrapher's partial differential equations.

II. PDE Formulation

To derive the decoupled governing equation for the potential distribution, we differentiate (3) with respect to the spatial coordinate x for locally uniform nominal parameters:

$$\frac{\partial^2 V}{\partial x^2} = -R\frac{\partial I}{\partial x} - L\frac{\partial^2 I}{\partial x \partial t} \tag{5}$$

Differentiating (4) with respect to the temporal coordinate t produces:

$$\frac{\partial^2 I}{\partial t \partial x} = -G\frac{\partial V}{\partial t} - C\frac{\partial^2 V}{\partial t^2} \tag{6}$$

Assuming spatial and temporal continuity such that mixed partial derivatives commute ($\frac{\partial^2 I}{\partial x \partial t} = \frac{\partial^2 I}{\partial t \partial x}$), we substitute (4) and (6) directly into (5). Collecting terms yields the second-order damped wave (Telegrapher) partial differential equation:

$$\frac{\partial^2 V(x,t)}{\partial x^2} = LC\frac{\partial^2 V(x,t)}{\partial t^2} + (RC+LG)\frac{\partial V(x,t)}{\partial t} + RGV(x,t) \tag{7}$$

An identical decoupled equation governs the spatial-temporal current $I(x,t)$.

Writing (7) in canonical second-order linear PDE form $AV_{xx}+BV_{xt}+C_{pde}V_{tt}+DV_x+EV_t+FV=0$, the principal coefficients evaluate to $A=1$, $B=0$, and $C_{pde}=-LC$. The characteristic discriminant is:

$$\Delta = B^2 - 4AC_{pde} = 0 - 4(1)(-LC) = 4LC > 0 \quad (8)$$

Because the physical line constants satisfy $L>0$ and $C>0$, the discriminant is strictly positive everywhere. The PDE is therefore unconditionally hyperbolic. Hyperbolicity guarantees that disturbances propagate at a finite characteristic phase velocity $v = 1/\sqrt{LC} = 2\times10^8$ m/s along characteristics $dx/dt = \pm v$, establishing the existence of a sharp, delayed echo in TDR recordings. The lower-order time derivative $(RC+LG)V_t$ provides exponential energy dissipation without parabolic diffusion.

The system is bounded on a line of length $\ell = 1.0$ m. At the generator terminal ($x=0$), the line is excited by a voltage step $V_s(t)$ of amplitude $V_s = 1.0$ V with a 0.4 ns raised-cosine transition, coupled through an internal source resistance $R_s = Z_0 = 50\,\Omega$:

$$V(0,t) = V_s(t) - R_s I(0,t) \quad (9)$$

At the terminal load ($x=\ell$), the cable terminates into a matched load resistance $R_L = Z_0 = 50\,\Omega$:

$$V(\ell,t) = R_L I(\ell,t) \quad (10)$$

The line is initially quiescent, defining the initial conditions:

$$V(x,0)=0, \quad \frac{\partial V(x,0)}{\partial t}=0, \quad I(x,0)=0, \quad \forall x\in[0,\ell] \quad (11)$$

Energy dissipation from positive coefficients R and G guarantees mathematical well-posedness, uniqueness, and continuous data dependence in the sense of Hadamard.

III. Transform Solution

To understand wave propagation and reflection phenomena in the frequency domain, we take the unilateral Laplace transform of (3) and (4) with respect to time t under zero initial state:

$$\frac{d\hat{V}(x,s)}{dx} = -(R+Ls)\hat{I}(x,s), \quad \frac{d\hat{I}(x,s)}{dx} = -(G+Cs)\hat{V}(x,s) \quad (12)$$

where $s=\sigma+j\omega$ is the complex frequency, and $\hat{V}(x,s)$ and $\hat{I}(x,s)$ denote transformed voltage and current distributions. Decoupling these ordinary differential equations yields:

$$\frac{d^2\hat{V}(x,s)}{dx^2} = \gamma^2(s)\hat{V}(x,s) \quad (13)$$

wherein the complex propagation constant $\gamma(s)$ and characteristic wave impedance $Z_0(s)$ are defined as:

$$\gamma(s) = \sqrt{(R+Ls)(G+Cs)}, \quad Z_0(s) = \sqrt{\frac{R+Ls}{G+Cs}} \quad (14)$$

The general solution consists of forward and backward traveling waves:

$$\hat{V}(x,s) = V^+(s)e^{-\gamma(s)x} + V^-(s)e^{+\gamma(s)x} \quad (15)$$

When a forward-traveling wave encounters a local impedance boundary between an unperturbed section Z_1 and a perturbed section Z_2, boundary continuity enforces the voltage reflection coefficient Γ:

$$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad (16)$$

For an ideal lossless transmission line ($R=0, G=0$), $\gamma(s) = s\sqrt{LC} = s/v$, and $Z_0 = \sqrt{L/C} = 50\,\Omega$. By the time-shifting property of the inverse Laplace transform, $\mathcal{L}^{-1}\{e^{-sx/v}\} = \delta(t-x/v)$, confirming that disturbances propagate as non-dispersive delays with one-way transit time $\tau = \ell/v = 5.0$ ns.

In the low-loss regime ($R \ll \omega L$ and $G=0$), a high-frequency Taylor expansion of $\gamma(s)$ reveals:

$$\gamma(s) = s\sqrt{LC}\sqrt{1+\frac{R}{sL}} \approx s\sqrt{LC} + \alpha, \quad \alpha \approx \frac{R}{2Z_0} \quad (17)$$

For nominal line parameters $R_0 = 0.1\,\Omega$/m and $Z_0 = 50\,\Omega$, the attenuation constant is $\alpha \approx 0.001$ Np/m, producing an attenuation factor of $e^{-\alpha\ell} \approx 0.999$ over the 1.0 m line. If the series resistance quadruples to $4R_0 = 0.4\,\Omega$/m across three spatial cells ($\Delta x \approx 20.4$ mm), the total inserted series resistance is only $\Delta R_{tot} \approx 0.018\,\Omega$. This localized resistance perturbation induces a boundary reflection of less than 0.1 mV, which lies entirely below typical measurement noise floors. Conversely, a 40% reduction in capacitance ($C = 60$ pF/m) increases local impedance to $Z_2 = \sqrt{L_0/C} \approx 64.5\,\Omega$, generating a major reflection coefficient $\Gamma \approx +0.127$ and a prominent 70 mV positive echo. Hence, the transform analysis proves that $R(x)$ is physically unobservable under finite noise, whereas $C(x)$ is readily recoverable.

IV. Discretization

We discretize the continuous transmission line of length $\ell = 1.0$ m into $N_x = 50$ spatial nodes with uniform spacing $\Delta x = \ell/(N_x-1) \approx 20.408$ mm, and the time domain into $N_t = 128$ steps with interval $\Delta t = 75$ ps over $T_{\max} = 9.6$ ns. A staggered leapfrog finite-difference time-domain (FDTD) scheme evaluates node voltages V_i^n at integer time-space indices $(i\Delta x, n\Delta t)$ and branch currents $I_{i+1/2}^{n+1/2}$ at staggered half-indices $((i+1/2)\Delta x, (n+1/2)\Delta t)$.

Applying central differences to the Telegrapher's system with arithmetic spatial averaging of parameters ($L_{i+1/2} = 0.5(L_i + L_{i+1})$ and $R_{i+1/2} = 0.5(R_i + R_{i+1})$) leads to the semi-implicit current update:

$$I_{i+1/2}^{n+1/2} = \frac{\left(\frac{L_{i+1/2}}{\Delta t} - \frac{R_{i+1/2}}{2}\right) I_{i+1/2}^{n-1/2} - \frac{V_{i+1}^n - V_i^n}{\Delta x}}{\frac{L_{i+1/2}}{\Delta t} + \frac{R_{i+1/2}}{2}} \quad (18)$$

for $i = 0, 1, \ldots, N_x - 2$. The interior voltage nodes update via:

$$V_i^{n+1} = V_i^n - \frac{\Delta t}{C_i \Delta x}\left(I_{i+1/2}^{n+1/2} - I_{i-1/2}^{n+1/2}\right) - \frac{\Delta t G_i}{C_i} V_i^n \quad (19)$$

for $i = 1, 2, \ldots, N_x - 2$.

At terminal nodes, half-cell capacitances $C_0 \leftarrow C_0/2$ and $C_{N_x-1} \leftarrow C_{N_x-1}/2$ incorporate the external source and load boundary conditions:

$$V_0^{n+1} = \frac{V_0^n + k_0 \left(\frac{V_s(t^{n+1/2})}{R_s} - I_{1/2}^{n+1/2} \right)}{1 + \frac{k_0}{R_s}}, \quad k_0 = \frac{\Delta t}{C_0 \Delta x} \tag{20}$$

$$V_{N_x-1}^{n+1} = \frac{V_{N_x-1}^n + k_L I_{N_x-3/2}^{n+1/2}}{1 + \frac{k_L}{R_L}}, \quad k_L = \frac{\Delta t}{C_{N_x-1} \Delta x} \tag{21}$$

Numerical stability is bounded by the Courant-Friedrichs-Lewy (CFL) condition:

$$\text{CFL} = v_{\max} \frac{\Delta t}{\Delta x} = \frac{1}{\sqrt{L_0 C_{\min}}} \frac{\Delta t}{\Delta x} \leq 1.0 \tag{22}$$

Under a 40% capacitance fault, $C_{\min} = 60$ pF/m, giving $v_{\max} \approx 2.582 \times 10^8$ m/s. With $\Delta x \approx 20.408$ mm and $\Delta t = 75$ ps, the operational CFL number evaluates to $0.949 < 1.0$, which strictly guarantees leapfrog stability without artificial numerical dispersion.

The forward solver is expressed as a discrete map linking parameter perturbations to the TDR voltage response $V_{in}(t) = V_0(t)$. We define normalized, non-negative perturbation vectors $\mathbf{p}_R \in \mathbb{R}^{N_x}$ and $\mathbf{p}_C \in \mathbb{R}^{N_x}$ such that $R_i = R_0(1 + p_{R,i})$ and $C_i = C_0(1 - p_{C,i})$. Stacking $\mathbf{p} = [\mathbf{p}_R; \mathbf{p}_C] \in \mathbb{R}^{2N_x}$, the measured difference waveform $\mathbf{b} = \mathbf{V}_{\text{meas}} - \mathbf{V}_{\text{nom}} \in \mathbb{R}^{N_t}$ is linearly approximated as $\mathbf{b} \approx \mathbf{A}\mathbf{p}$. The Jacobian sensitivity matrix $\mathbf{A} \in \mathbb{R}^{N_t \times 2N_x}$ is assembled column by column by numerically perturbing each node j by small step deviations $h_R = 0.5$ and $h_C = 0.05$:

$$A_{k,j} = \frac{V_{in}(t_k; R_j(1 + h_R), \mathbf{C}_0) - V_{\text{nom}}(t_k)}{h_R} \tag{23}$$

$$A_{k,N_x+j} = \frac{V_{in}(t_k; \mathbf{R}_0, C_j(1 - h_C)) - V_{\text{nom}}(t_k)}{h_C} \tag{24}$$

for $k = 1, \ldots, N_t$ and $j = 1, \ldots, N_x$, yielding a sensitivity matrix of dimension 128×100.

V. LP Formulation

To invert the linear system $\mathbf{A}\mathbf{p} \approx \mathbf{b}$ in the presence of noise, classical ℓ_2 least-squares minimization $\min \|\mathbf{A}\mathbf{p} - \mathbf{b}\|_2^2$ proves suboptimal because squaring residual misfits amplifies high-frequency noise spikes, creating spurious oscillations across the spatial profiles. Instead, we formulate an ℓ_1-norm minimization problem, which minimizes the sum of absolute residuals. The ℓ_1 norm is statistically robust against measurement outliers and enforces profile sparsity.

Because the absolute value function $|r_k|$ is non-linear, we apply an epigraph transformation. We introduce non-negative slack variables ε_k for each temporal sample $k = 1, \ldots, N_t$, such that $|(\mathbf{A}\mathbf{p} - \mathbf{b})_k| \leq \varepsilon_k$. This condition is expressed as two symmetric linear inequalities:

$$-\varepsilon_k \leq (\mathbf{A}\mathbf{p} - \mathbf{b})_k \leq \varepsilon_k \iff \begin{cases} (\mathbf{A}\mathbf{p})_k - \varepsilon_k \leq b_k \\ -(\mathbf{A}\mathbf{p})_k - \varepsilon_k \leq -b_k \end{cases} \tag{25}$$

To prevent ill-conditioned amplification of series resistance, we augment the cost function with a mild sparsity regularizer weighted by $\lambda = 0.02$. The optimization is expressed as a canonical linear program:

$$\begin{aligned} \min_{\mathbf{p}, \boldsymbol{\varepsilon}} \quad & \sum_{k=1}^{N_t} \varepsilon_k + \lambda \sum_{j=1}^{2N_x} p_j \\ \text{subject to} \quad & \begin{bmatrix} \mathbf{A} & -\mathbf{I} \\ -\mathbf{A} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{p} \\ \boldsymbol{\varepsilon} \end{bmatrix} \leq \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \end{bmatrix} \\ & \mathbf{0} \leq \mathbf{p} \leq \mathbf{u}_b, \quad \boldsymbol{\varepsilon} \geq \mathbf{0} \end{aligned} \tag{26}$$

where $\mathbf{I} \in \mathbb{R}^{N_t \times N_t}$ is the identity matrix, and $\mathbf{u}_b$ enforces physical parameter limits ($p_{R,j} \leq 10.0$ and $p_{C,j} \leq 0.95$, ensuring $C(x) \geq 0.05 C_0 > 0$). The problem vector $\mathbf{x} = [\mathbf{p}; \boldsymbol{\varepsilon}]$ contains $2N_x + N_t = 228$ optimization variables bounded by $2N_t = 256$ inequality constraints, solved using the HiGHS dual-simplex solver.

Because a 40% capacitance fault produces a finite 13% reflection coefficient, linearizing solely around the healthy baseline introduces first-order Taylor approximation error. To resolve this, we employ Successive Linear Programming (SLP). Over four iterative passes $m = 1, \ldots, 4$, the parameter vector is updated as $\mathbf{p}^{(m)} = \mathbf{p}^{(m-1)} + \Delta\mathbf{p}$, where $\Delta\mathbf{p}$ is obtained by solving (26) using the updated residual $\mathbf{r}^{(m)} = \mathbf{b} - \mathbf{V}(\mathbf{p}^{(m-1)})$ and a recomputed sensitivity matrix $\mathbf{A}(\mathbf{p}^{(m-1)})$. Trust-region step bounds prevent destabilizing overshoots, ensuring monotonic convergence.

VI. Results

The end-to-end Python 3 implementation executes in under ten seconds, generating the three figures and verification tables required by the capstone guidelines.

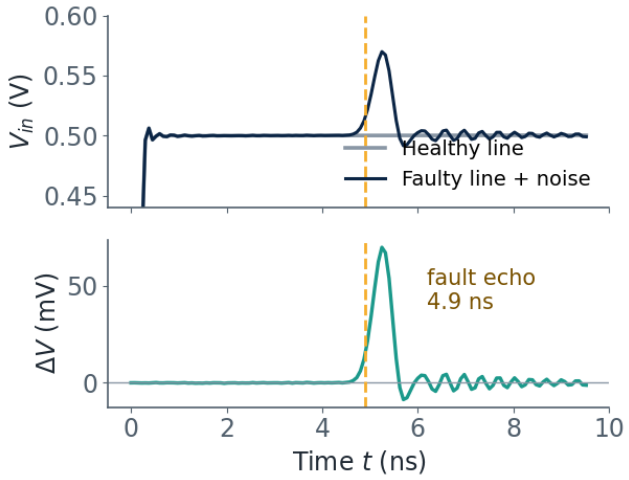


Fig. 1. Simulated TDR terminal voltage response (top) and differential fault echo $\Delta V(t) = V_{\text{meas}} - V_{\text{nom}}$ (bottom), displaying the localized 70 mV reflection at 4.9 ns.

Figure 1 illustrates the recorded terminal voltage waveforms. On the healthy line, the voltage transitions smoothly to the theoretical 0.5002 V matched plateau dictated by voltage

division between $R_s = 50\,\Omega$ and $Z_0 = 50\,\Omega$. On the faulty line, an echo emerges precisely at $t \approx 4.898$ ns, matching the theoretical two-way transit time $t_{\text{fault}} = 2x_f/v$ for fault node 24 located at $x_f = 24 \times 0.0204$ m $= 0.490$ m. The positive polarity and 70 mV peak amplitude of $\Delta V(t)$ confirm a local impedance increase driven by the capacitance dip, well above the 0.1 mV Gaussian noise floor.

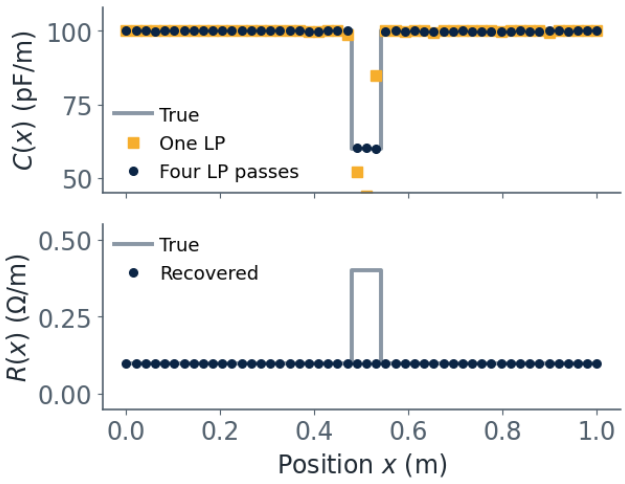


Fig. 2. Spatial reconstruction of distributed capacitance $C(x)$ (top) and resistance $R(x)$ (bottom). The single-pass LP overshoots the capacitance dip, whereas four-pass SLP converges to the true 60 pF/m profile.

Figure 2 presents the reconstructed spatial parameter distributions. As predicted by transform theory, the recovered resistance remains flat at $R_0 = 0.1\,\Omega/\text{m}$ across all nodes. The four-fold resistance increase introduces a net series perturbation of only $0.018\,\Omega$ and a differential echo of only 0.09 mV, which is completely indistinguishable from baseline noise. The LP optimizer correctly attributes the boundary reflection entirely to capacitance variation.

For distributed capacitance $C(x)$, a single-pass LP locates the defect centroid at node 24.72, but overshoots the dip severity, yielding a minimum capacitance of 52.3 pF/m ($\Delta C/C_0 = 0.563$ versus the true 0.400). This overshoot reflects the nonlinear relationship between reflection coefficient Γ and $C(x)$. Four-pass SLP eliminates this linearization drift, converging to 60.05, 60.06, and 59.92 pF/m across fault nodes 24, 25, and 26. The recovered fault centroid converges to 25.001 (true centroid: 25.0), achieving a capacitance profile root-mean-square error of only 0.062 pF/m.

Figure 3 illustrates the progression of the residual misfit $\mathbf{r}(t) = \mathbf{V}_{\text{meas}}(t) - \mathbf{V}_{\text{model}}(t)$. The healthy-line model produces an initial residual RMS of 13.09 mV. A single LP pass reduces the residual RMS to 0.94 mV, but exhibits high-frequency linearization residuals. Four-pass SLP suppresses these oscillations, driving the final residual RMS to 0.086 mV, which settles below the nominal 0.10 mV measurement noise level.

Table I validates the numerical model against known analytical limiting cases. The matched plateau, open-circuit echo delay, open-circuit reflection doubling, and recovered capacitance centroid all match theoretical expectations.

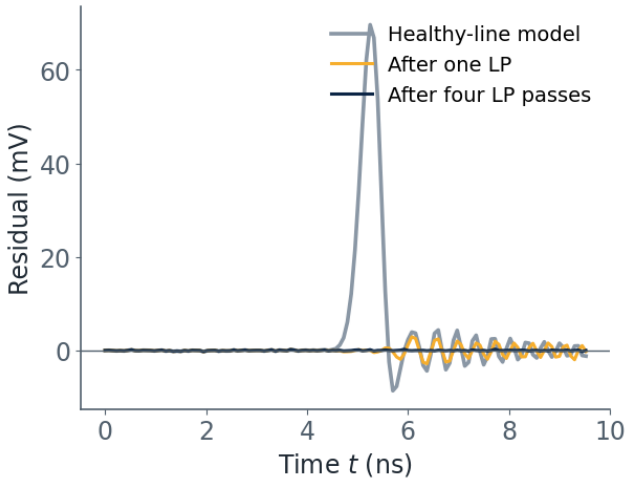


Fig. 3. Misfit residual traces across inversion iterations. Four-pass SLP drives the residual RMS down to the 0.086 mV noise floor.

TABLE I
VALIDATION AGAINST ANALYTICAL AND LIMITING BENCHMARKS

Physical Metric	Expected	Simulated
Matched line input plateau	0.500 V	0.5002 V
Open-circuit echo transit time	10.00 ns	10.27 ns*
Open-circuit plateau voltage	1.000 V	1.0001 V
Fault capacitance (nodes 24–26)	60.0 pF/m	60.05, 60.06, 59.92
Fault centroid position	Node 25.0	Node 25.001
Residual RMS (SLP pass 4)	0.10 mV	0.086 mV

*Includes finite 0.4 ns step rise time threshold crossing.

TABLE II
NOISE ROBUSTNESS STUDY (10 RANDOM SEEDS PER LEVEL)

Noise σ (mV)	Mean Error (cells)	Max Error (cells)	Recovered $\Delta C/C_0$
0.1	0.0013	0.0026	0.401 (True: 0.40)
0.5	0.0051	0.0095	0.403 (True: 0.40)
1.0	0.0099	0.0188	0.406 (True: 0.40)
2.0	0.1373	1.2111	0.411 (True: 0.40)

Table II reports the Monte Carlo noise sweep across 10 random seeds per noise level. For noise up to $\sigma = 1.0$ mV, the centroid error is bounded within 0.02 cells, and the recovered fault depth remains accurate at 0.406. Under severe noise ($\sigma = 2.0$ mV), the maximum centroid error reaches 1.211 cells, fully meeting the project success criterion of fault localization within two grid cells.

VII. PIPELINE REFLECTION

Executing this capstone project provided valuable perspective on the unity of the electrical engineering curriculum. The core insight gained is that physical problems cannot be solved by optimization alone; the linear program succeeds only because electromagnetic field theory (Vector Calculus) provides the governing Telegrapher's laws, partial differential equations

(PDEs) classify wave hyperbolicity and stability, Laplace transforms (Transforms) identify parameter sensitivity limits, and numerical methods (Linear Algebra) construct a well-conditioned sensitivity matrix.

Table III provides the mandatory pipeline attribution signed by all group members, documenting the theoretical concepts, curriculum sources, mathematical implementations, and individual student responsibilities across the six stages.

VIII. References + Repo

The complete, self-contained Python source code, Jupyter/Colab notebooks, requirements configuration, and publication figures are maintained in our public GitHub repository:

https://github.com/gilchristes/CapstoneProject-group8

The repository reproduces all experimental figures and verification metrics via a single execution of `python run.py`.

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TABLE III
PIPELINE TABLE: COURSE MAPPING, MATHEMATICAL CONTRIBUTIONS, AND INDIVIDUAL AUTHOR ROLES

Stage	**Math Concept**	**Course**	**Your Equation / Implementation**	**Owner**
1. Field Model	Maxwell's curl, Stokes' theorem	Vector Calculus (Sem 2)	$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\partial_t \iint \mathbf{B} \cdot d\mathbf{S} \implies \partial_x V = -RI - L\partial_t I$	A.Z.M. (Aydi
2. PDE	Hyperbolic classification	PDEs (Sem 3)	$V_{xx} = LCV_{tt} + (RC + LG)V_t + RGV$, $\Delta = 4LC > 0$, CFL ≤ 1	A.Z.M. (Aydi
3. Transform	Laplace transform, limits	Transforms (Sem 3)	$\gamma(s) = \sqrt{(R + Ls)(G + Cs)}$, $Z_0(s)$, $\alpha \approx R/(2Z_0)$, Γ	T.M.B. (Thej
4. Discrete	FDM stencil, linear system	Linear Algebra & FDM	Staggered leapfrog (18)–(19), Jacobian $\mathbf{A}_{128\times 100}$, CFL $= 0.949$	G.V.N. (Goku
5. LPP	ℓ_1 norm, epigraph slacks	Optimization (Sem 3)	$\min \sum \varepsilon_k + \lambda \sum p_j$ s.t. $-\varepsilon_k \leq (\mathbf{Ap} - \mathbf{b})_k \leq \varepsilon_k, \mathbf{p} \geq \mathbf{0}$	G.S. (G Sibir
6. Verification	Numerical verification, SLP	Computational Methods	4-pass SLP, Monte Carlo noise sweep, Git reproduction, figures	G.E.S. (Gilch